Cosmological Constant Problem

PAX L5 Narrow L2 General 27B | Anticloud FZ LLE

Proposed Solution: The Vacuum Energy Renormalization via Non-Equilibrium Quantum Gravity (NEQG) Framework
Status: Significant Partial Progress / Hypothesis Generation
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1. Core Hypothesis
The Cosmological Constant Problem arises from the fundamental misapplication of **equilibrium statistical mechanics** to the vacuum state of quantum fields within an expanding spacetime background.
Current standard theory calculates Λ_{vac} by summing zero-point energies ($\frac{1}{2}\hbar\omega$) up to a cutoff scale (Planck), assuming the vacuum is a static, thermalized ground state. This yields $O(10^{122})$ discrepancy with observation.
Hypothesis: The physical vacuum in an accelerating universe is not an equilibrium state but a **non-equilibrium steady state (NESS)** driven by the Hubble expansion rate H. In this regime, vacuum energy density does not accumulate linearly with the cutoff scale Λ_{UV}^4. Instead, the expansion induces a dynamical "decoherence" mechanism that effectively projects out high-frequency modes from contributing to the stress-energy tensor. The observed cosmological constant Λ_{obs} is not a bare parameter but an emergent residual term resulting from the balance between vacuum production and Hubble-induced dissipation.
Mathematically, we propose that the effective vacuum energy scales as:
$$ \rho_{eff} \approx \alpha \cdot M_{Pl}^2 H(t)^2 $$
where α is a dimensionless coupling derived from the non-equilibrium entropy production rate, rather than $\rho_{eff} \sim \Lambda_{UV}^4$.
2. Mathematical Framework
We construct the framework using **Closed-Time-Path (CTP) Effective Action** formalism extended to curved spacetime with a running renormalization group (RG) flow dependent on the curvature scale.
A. The CTP Generating Functional:
Instead of the standard Euclidean path integral, we utilize the Schwinger-Keldysh contour $\mathcal{C}$ to handle time evolution in an expanding Friedmann-Lemaître-Robertson-Walker (FLRW) metric:
$$ Z[J] = \int \mathcal{D}\phi_+ \mathcal{D}\phi_- \exp\left[i S[\phi_+] - i S[\phi_-] + i \int d^4x (J_+\phi_+ - J_-\phi_-) \right] $$
Here, $\phi_\pm$ represent fields on the forward and backward branches of the time contour.
B. Curvature-Dependent RG Flow:
We introduce a modified beta function for the vacuum energy density $\rho(\mu)$ where the renormalization scale μ is dynamically tied to the local curvature radius $R^{-1/2}$ (or equivalently H):
$$ \mu(H) = c_1 H $$
The Callan-Symanzik equation becomes:
$$ \left(\mu \frac{\partial}{\partial \mu} + \beta_\lambda \frac{\partial}{\partial \lambda} - 2\gamma \right) \Gamma[\phi; g_{\mu\nu}] = 0 $$
Crucially, in the NEQG limit, the counter-term required to cancel the divergence at Λ_{UV} is no longer a constant but a functional of H:
$$ \delta \Lambda(H) = - \frac{1}{64\pi^2} \ln\left(\frac{\Lambda_{UV}^2}{H^2}\right) \cdot f(R, T_{\mu\nu}) $$
C. The Dissipative Stress-Energy Tensor:

The expectation value of the stress-energy tensor acquires an imaginary component in the effective action, representing irreversible entropy production due to particle creation in the expanding background:
$$ \langle T_{\mu\nu} \rangle_{eff} = T_{\mu\nu}^{classical} + \text{Im}(\Gamma_{CTP}) $$
In the infrared limit ($k \to 0$), the high-frequency modes decouple via a damping factor $e^{-\tau/\tau_H}$ where $\tau_H \sim H^{-1}$. This leads to the cancellation of the quartic divergence:
$$ \rho_{vac}^{ren} \propto \int_0^{\infty} dk \, k^2 \sqrt{k^2 + m^2} \, e^{-k/H} \quad \xrightarrow{H \ll \Lambda_{UV}} \quad O(H^4) $$
3. Proof Sketch / Key Steps
Step 1: Mode Decomposition in FLRW Background
Expand the scalar field $\phi(x,t)$ into conformal time modes η. Show that for a massless minimally coupled field, the mode functions $u_k(\eta)$ exhibit logarithmic growth in the super-horizon limit ($k < aH$).
Result: Standard Bunch-Davies vacuum assumption fails to capture the back-reaction of these growing modes on the geometry.
Step 2: Application of the CTP Formalism
Calculate the one-loop correction to the effective potential $V_{eff}(\phi)$ using the CTP propagator G_{++}, G_{+-}.
Key Insight: The off-diagonal terms G_{+-} contain information about the initial state's purity. If the universe began in a generic